

Applied Generative AI

Session 2 – Assignment

Instructions

- Read the complete Session 2 notes before attempting this assignment.
- Answer all questions in your own words.
- Use AI tools only for learning and research. Your explanations should reflect your own understanding.
- Submit your work as a single PDF document.

Section A – Conceptual Understanding

1. Explain the complete journey of a prompt inside a Large Language Model.

Use the following pipeline in your explanation:

User Prompt → Tokenizer → Tokens → Token IDs → Embeddings → Transformer → Attention →
Next Token Prediction → Response

2. Explain the following terms using your own words and a real-world analogy.

- Token
- Embedding
- Attention
- Transformer
- Context Window
- Hallucination

3. Differentiate between the following:

- Training vs Inference
- Token IDs vs Embeddings
- Zero-shot vs One-shot vs Few-shot Prompting

Present your answer in a comparison table.

Section B – Practical Activities

4. Prompt Engineering Exercise

Write three prompts for the same task:

Explain Machine Learning.

- A basic prompt
- An improved prompt
- A professional prompt using the **RTCC Framework** (Role, Task, Context, Constraints)

Compare the outputs and write your observations.

5. Compare AI Models

Use any **two** AI models (e.g., ChatGPT, Gemini, Claude, Copilot).

Ask both models the same prompt:

Explain Artificial Intelligence to a 12-year-old.

Compare their responses based on:

- Clarity
- Accuracy
- Examples
- Writing Style
- Overall Quality

Write a short conclusion.

6. Explore Tokenization

Choose a paragraph of approximately 100 words.

Use an AI tokenizer or an online tokenizer to observe how the paragraph is divided into tokens.

Write your observations about:

- Number of words
- Approximate number of tokens
- Why tokens are different from words

Section C – Research Activity

7. Research and compare the following Large Language Models.

- GPT
- Gemini
- Claude
- Llama

Create a comparison table including:

- Developer
- Open or Closed
- Primary Strengths
- Typical Use Cases

Section D – Reflection

8. Reflection

Write a short note (300–500 words) on:

"Why does changing the prompt change the AI's response?"

Support your answer with examples from your practical activities.

Submission Checklist

Before submitting, ensure that your assignment includes:

- Your Name
- Email Address
- Date of Submission
- All answers in a single PDF document
- Screenshots wherever applicable

For submission, create a copy of this document and complete it here:

<https://docs.google.com/document/d/1sGofmqIJMvkPGWWkY3iCMGuEudfh8J0Vo2cXztNHJBo/edit?usp=sharing>